STAFF REVIEW · H446 041 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Sends a value back to its caller

2. Repeats while a test holds

3. It halves the remaining range

4. B

I Do — source worked steps

1. Identify both variables — Program.score is a class-level static field, initially 10. Main declares its own local score, initially 3.

2. Resolve each name — The unqualified score inside Main refers to the local. Program.score explicitly selects the class field; the field has not changed.

3. Explain the output and scope — The output is 3 then 10. Another method cannot directly access Main's local score. Restricting scope reduces unintended changes to shared state.

We Do — guided CHECK

1. Check your thinking
The parameter n with value 4.

2. Discuss, then respond
Why avoid excessive global state?
Check your thinking
Changes from one part of the program can affect another.

You Do — source mark schemes

1. They limit scope and reduce unintended interference.

2. No.

3. A declaration in an inner scope uses the same name as one in an outer scope, hiding it there.

Exit Ticket — source answer

1. False. A local variable is visible only within its defined scope.